Practice Exam II

Math 1320 001 Summer 2014
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July 25th 2014

Name and Unid: _____
by writing my name I swear by the honor code

Read all of the following information before starting the exam:
Note this is the practice exam ignore point values on problems there are extras but the exam will look similar

- Show all work, clearly and in order, if you want to get full credit. I reserve the right to take off points if I cannot see how you arrived at your answer (even if your final answer is correct).
- Justify your answers algebraically whenever possible to ensure full credit. Sketch all relevant graphs and explain all relevant mathematics.
- Circle or otherwise indicate your final answers.
- You MAY NOT use a calculator or notes. You may use the formulas provided.
- This test has 11 problems and one bonus problem and is worth 100 points, plus some extra credit at the end. It is your responsibility to make sure that you have all of the pages!
- Good luck!

Good Luck!
Here are some Formulas and rules for you.

1. (*5 points*)
Find the distance between the points $(1, 4, -6)$ and $(2, 3, 7)$

2. (*10 points*)
Show that the equation represents a sphere, find it's center and radius. $2x^2 + y^2 + z^2 + 8x - 6y + 2z = 0$

3. (*10 points*) Find $\vec{a}+\vec{b}, \vec{a}+\vec{b}, |\vec{a}|, |\vec{a}+\vec{b}|$ for the vectors $\vec{a} =< 1, 4, -2 >, \vec{b} =< 2, 5, -1 >$

4. (*5 points*) Find a unit vector in the same direction as $\vec{v} =< -2, 4, 2 >$. Also find a vector in the same direction but with length 6.

5. (*10 points*) Two forces are acting on a particle in the x-y plane given by: $F_1 =< 2, 6 >$ and $F_2 =< 1, -2 >$. Draw these two force vectors. What is the angle between the vectors? Find the total force vector and determine what angle will the particle move relative to the x-axis.

6. (*10 points*) Determine which of the following pairs of vectors are orthogonal.

- $< 1, 4, 6 >, < -1, -4, 6 >$
- $< 1, 4 >, < 4, -1 >$
- $< 1, -2, 1 >, < -1, -1, -1 >$
- $< 1, 0 >, < -1, 0 >$

7. (*5 points*) Find the cross product between $\vec{a} =< 6, 0, -2 >$ and $\vec{b} =< 0, 4, 2 >$ and verify it is orthogonal to both $\vec{a}$ and $\vec{b}$

8. (*15 points*)

a. (*5 pts*) Find the symmetric, parametric , and vector equations of the line between the points $(6, 1, 2)$ and $(2, 3, 5)$

b. (*10 pts*) Find the equation of the plane that contains the points $(1, 2, 4)$, $(4, -2, 3)$ and $(5, 2, 1)$

9. (*10 points*) Find the distance from the point $(4, 1, -2)$ to the line $\vec{r}(t) = < 1+t, 3-2t, 4-3t >$

10. (*20 points*) Consider the function $f(x,y) = ln(\sqrt{4-x^2+y^2})$

a. (*5 pts*) Find the domain of $f(x,y)$

b. (*5 pts*) Sketch the domain of $f(x,y)$

11. (*10 points*) Consider the function $f(x, y) = \frac{x^2}{4} + y^2$ Sketch level sets for $z = 0, 1, 2, 3$ Use those to sketch a graph of $f(x, y)$

12. (*15 points*) Do the following coordinate changes and plots

a. (*5 pts*) Change $(1, \sqrt{3}, 2\sqrt{3})$ from Rectangular to spherical coordinates.

b. (*5 pts*) Plot the point given in cylindrical coordinates as $(2, \frac{\pi}{3}, 1)$ Then convert it to rectangular coordinates.

c. (*5 pts*) Plot the point in spherical coordinates given by $(2, \frac{\pi}{2}, \frac{\pi}{4})$ Then convert it to rectangular coordinates.

13. (*10 points*) Make a sketch of the curve given by the vector equation $\vec{r}(t) = < 2cos(t), 2sin(t), 2t >$

14. (*10 points*) Find the velocity ,acceleration and speed of a particle whose position is given by $\vec{r}(t) =< 1 + t^2, t^2, sin(\pi t) >$ at $t = 1$

15. (*10 points*) A particle is traveling in the x,y plane with velocity $\vec{v}(t) =< -2sin(t), 2cos(t) >$ how fast is the particle moving? How far did the particle move from time $t = \pi/2$ to $t = \pi$?

16. (*10 points*) Find the following vectors using the vector function $\vec{r}(t) =< 2sin(t), 5t, 2cos(t) >$

a. (*5 pts*) Find the unit tangent vector $\vec{T}(t)$

b. (*5 pts*) Find the unit Normal vector $\vec{N}(t)$

c. (*5 pts*) Find the curvature κ

17. (*10 points*) Consider the function $f(x, y) = x^3 - y$

a. (*5 pts*) Draw a contour map of the of $f(x, y)$

b. (*5 pts*) Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$

c. (*5 pts*) Find $\frac{\partial^2 f}{\partial y \partial x}$

18. (*10 points*)
Suppose $f(x,y,z) = 3x^2 + 4xy^4 + 3xyz$ and $x = scos(3t)$, $y = sy^3$, $z = tcos(s)$

a. (*5 pts*) Find $\frac{\partial f}{\partial t}$

b. (*5 pts*) Find $\frac{\partial f}{\partial s}$

c. (*5 pts*) Find $\frac{\partial z}{\partial x}$

19. (*10 points*)
Find the equation of the tangent plane to the surface $f(x, y) = x^2 + xy^3$ at the point $(1, 1, 2)$

20. (*5 points*) (Bonus all or nothing) Find a solution (by Guessing or looking one up before hand) to Laplace's equation $f_{xx} + f_{yy} = 0$ State your solution and show that is solves the PDE.

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